



Summary

Sample	Spoligotype (octal)	SB / SIT	Species	Lineage	Pattern (spacers 1 to 43)	Status
AF2122_97	664073777777600	SB0140 SIT683	M. bovis	BOV		OK
NC_002945	664073777777600	SB0140 SIT683	M. bovis	BOV		OK
H37Rv	777777477760771	SIT451	M. tuberculosis	4.9		OK

Review

Reviewed by		Date		Signature	
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Reads per spacer

Blue: present (count ≥ 5). Orange: called absent but seen in some reads.

AF2122_97 OK**Species and lineage**

Species	M. bovis
Lineage	BOV · M. bovis, M. caprae, M. orygis · typical spoligotypes: BOV_2, BOV_1
Livestock lineage	La1.8.1 · Eu1 (Zwyer et al. 2021)
MTBC DNA	median 80.5 reads per control region, 100% of the control regions found; about 100% of the reads are MTBC

Region	H37Rv region	Usually deleted in	Segments found	Depth vs control	Result
RD1	4,350,251-4,359,740	BCG (in part: M. microti, Dassie bacillus)	20 of 20	0.80	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	0 of 20	0.00	– deleted
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	0 of 20	0.00	– deleted
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	0 of 8	0.00	– deleted
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	0 of 20	0.00	– deleted

+ / –: DNA of the region present in / absent from the sample, from 100 bp segments inside the region (not from amplicon sizes). See "Definitions and methods".

Lineage SNPs (Coll et al. 2014)

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads	Result
BOV_AFRI	1,882,180	Rv1662	62	0	detected
BOV	2,831,482	Rv2515c	56	0	detected

Result: detected, the lineage allele is in at least 80% of the reads covering the SNP (at least 3 reads); mixed, both alleles in at least 15% of 10 or more reads; -, not detected.

Livestock lineage SNPs (Zwyer et al. 2021)

Group	Name	SNPs covered	Derived allele	Both alleles	Reads with derived allele	Other reads	Result
La1/La2	-	5	5	0	315	2	detected
La1	M. bovis	4	4	0	269	0	detected
La1.8	Eu1, unknown6, unknown7, unknown8	4	4	0	305	0	detected
La1.8.1	Eu1	5	5	0	488	0	detected

SNPs covered: with at least 3 reads. Derived allele: SNPs with the derived allele in at least 80% of their reads. Both alleles: SNPs with each allele in at least 15% of 10 or more reads. Result: detected, at least 2 SNPs with the derived allele; mixed, at least 2 SNPs with both alleles; -, not detected.

Contigs per spacer

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

NC_002945 OK**Species and lineage**

Species	M. bovis
Lineage	BOV · M. bovis, M. caprae, M. orygis · typical spoligotypes: BOV_2, BOV_1
Livestock lineage	La1.8.1 · Eu1 (Zwyer et al. 2021)
MTBC DNA	median 1 contig per control region, 100% of the control regions found

Region	H37Rv region	Usually deleted in	Segments found	Depth vs control	Result
RD1	4,350,251-4,359,740	BCG (in part: M. microti, Dassie bacillus)	20 of 20	1.00	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	0 of 20	0.00	– deleted
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	0 of 20	0.00	– deleted
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	0 of 8	0.00	– deleted
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	0 of 20	0.00	– deleted

+ / –: DNA of the region present in / absent from the sample, from 100 bp segments inside the region (not from amplicon sizes). See "Definitions and methods".

Lineage SNPs (Coll et al. 2014)

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads	Result
BOV_AFRI	1,882,180	Rv1662	1	0	detected
BOV	2,831,482	Rv2515c	1	0	detected

Result: detected, the lineage allele is in at least 80% of the contigs covering the SNP (at least 1 contig); mixed, both alleles in at least 15% of 10 or more reads; -, not detected.

Livestock lineage SNPs (Zwyer et al. 2021)

Group	Name	SNPs covered	Derived allele	Both alleles	Reads with derived allele	Other reads	Result
La1/La2	-	5	5	0	5	0	detected
La1	M. bovis	4	4	0	4	0	detected
La1.8	Eu1, unknown6, unknown7, unknown8	4	4	0	4	0	detected
La1.8.1	Eu1	5	5	0	5	0	detected

SNPs covered: with at least 1 contig. Derived allele: SNPs with the derived allele in at least 80% of their contigs. Both alleles: SNPs with each allele in at least 15% of 10 or more reads. Result: detected, at least 2 SNPs with the derived allele; mixed, at least 2 SNPs with both alleles; -, not detected.

Contigs per spacer

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

H37Rv OK**Species and lineage**

Species	M. tuberculosis
Lineage	4.9 · Euro-American (H37Rv-like) · typical spoligotypes: T1
MTBC DNA	median 1 contig per control region, 100% of the control regions found

Region	H37Rv region	Usually deleted in	Segments found	Depth vs control	Result
RD1	4,350,251-4,359,740	BCG (in part: M. microti, Dassie bacillus)	20 of 20	1.00	+ present
RD4	1,696,001-1,708,740	M. bovis, BCG (some M. canettii)	20 of 20	1.00	+ present
RD7	2,208,001-2,220,740	M. africanum lineage 6, animal lineages	20 of 20	1.00	+ present
RD9	2,330,051-2,332,140	M. africanum (lineages 5, 6), animal lineages	8 of 8	1.00	+ present
RD12	3,485,101-3,487,540	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)	20 of 20	1.00	+ present

+ / -: DNA of the region present in / absent from the sample, from 100 bp segments inside the region (not from amplicon sizes). See "Definitions and methods".

Lineage SNPs (Coll et al. 2014)

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads	Result
4	931,123	Rv0835	1	0	detected
4.9	1,759,252	Rv1552	1	0	detected

Result: detected, the lineage allele is in at least 80% of the contigs covering the SNP (at least 1 contig); mixed, both alleles in at least 15% of 10 or more reads; -, not detected.

Definitions and methods

Spoligotype	The presence (1, filled square) or absence (0, empty square) of the 43 spacers of the direct repeat (DR) locus, written as a 43-digit binary pattern, a 15-digit octal code (Dale et al. 2001) or a hexadecimal code. These codes are universal: they identify the pattern itself, and are the ones to use to exchange spoligotypes.
SB number	The name of a pattern in the Mbovis.org database of spoligotypes of the RD9-deleted lineages (M. bovis and other animal-adapted lineages). "Not in Mbovis.org" only means that this database has no name for the pattern, as for all M. tuberculosis patterns: it does not mean that the spoligotype is invalid or new.
SIT	Shared international type of the SITVIT2 database (Institut Pasteur de Guadeloupe), from the SITVIT2 patterns published with SpolLineages (9,656 patterns, 3,850 SITs, 2022 list). "Orphan": a SITVIT2 pattern without SIT. "Not in SITVIT2 list": the pattern is not in this list, which does not include the SITs created since 2022. SITVIT2 lacks many patterns of the animal-adapted lineages: only 603 of the 1,976 Mbovis.org patterns have a SIT in this list. For these lineages, the SB number is the reference name, and no closest SIT is given for a pattern with an SB number.
Spoligotype family	SITVIT2 family of the pattern (e.g. Beijing, LAM3, BOV_1), and the spoligotype families typical of the lineage (Coll et al. 2014). Families are labels of patterns, not phylogenetic lineages.
Lineage	From the 62 SNPs of the barcode of Coll et al. (2014): the reads carrying each allele are counted with exact 31-mers; a lineage is called when at least 80% of the reads (at least 3, or 1 contig) carry its allele.
Livestock lineage	Lineage of the livestock-associated MTBC after Zwyrer et al. (2021): La1 (M. bovis), La2 (M. caprae), La3 (M. orygis), and the La1 sublineages La1.1 to La1.8, with their former names (e.g. La1.8.1: clonal complex Eu1; BCG belongs to La1.2). From 88 of the 89 marker SNPs of the extended data of the paper (4 or 5 per group), counted as above; a group is called when at least 2 of its SNPs carry the derived allele, as in the KvarQ test suite of the paper. La2 and La3 tell M. caprae from M. orygis, which have the same RD profile.
L1 sublineage	Sublineage of lineage 1 (East-African-Indian) after Netikul et al. (2022), in the revised nomenclature of lineage 1 (L1.1 to L1.3, down to e.g. L1.1.1.10), from the 1,835 sublineage-specific SNPs of the paper (4 to 224 per sublineage), counted as above; a sublineage is called when at least 2 of its SNPs, and at least half of those covered, carry the derived allele, and its parent group is called. The names of Coll et al. correspond to these groups: 1.2.1 is L1.2 (L1.2.1 and L1.2.2), 1.2.2 is L1.3, and 1.1.2 is L1.1.2.2 (L1.1.2.1 strains are 1.1).

Regions of difference

Regions of difference (RD) are determined in silico, from the sequencing data, without PCR. Each region is the part of the H37Rv genome missing from M. bovis AF2122/97 (RD4, RD7, RD9, RD12) or from BCG Pasteur (RD1), and is represented by 100 bp segments inside it (table below), chosen to be found in all the MTBC genomes that have the region and in none of those lacking it or of six non-tuberculous mycobacteria. A segment is found when Seal (25-mers, 1 mismatch) finds it in the reads at a depth of at least 5% of the depth of MTBC-specific control regions, or in the assembly. A region is present when all its segments are found (in reads, up to 10% of them, at least 1, may be missing: low depth); deleted when at most 10% of its segments (at least 1: stray reads) are found; partially deleted in between, with the H37Rv coordinates of the missing segments; present at reduced depth when, in reads, its segments are found at less than 50% of the control depth (mixed sample?).

Region	H37Rv region (NC_000962.3)	Segments	Usually deleted in
RD1	4,350,251-4,359,740 (9,490 bp)	20	BCG (in part: M. microti, Dassie bacillus)
RD4	1,696,001-1,708,740 (12,740 bp)	20	M. bovis, BCG (some M. canettii)
RD7	2,208,001-2,220,740 (12,740 bp)	20	M. africanum lineage 6, animal lineages
RD9	2,330,051-2,332,140 (2,090 bp)	8	M. africanum (lineages 5, 6), animal lineages
RD12	3,485,101-3,487,540 (2,440 bp)	20	M. bovis, BCG, M. caprae, M. orygis (some M. canettii)

The + and – of the RD profile therefore mean that the DNA of the region is present or absent, like a PCR with primers inside the region (amplification = present). spoligotyper does not measure amplicon sizes or deletion junctions: assays that distinguish RDs by the size of an amplicon spanning the region may report partial or strain-specific deletions differently. A partially deleted region counts as + when at least half of its segments are found: for example, M. microti lacks the part of RD1 inside its own RD1mic deletion (reported as partially deleted) and is RD1 + in the RD profile, as in the classical RD PCR scheme. When the profile then matches no species, but would with a partially deleted region present, that region counts as present for the species (a deletion of one strain), as the warning says. The species is read from the RD profile, refined with the lineage SNPs and the spacers.

Spacers and codes

Spacers are counted with Seal (BBTools): each of the 43 spacers is searched as a single 25-mer on both strands, allowing 1 mismatch (k=25, hdist=1, rcomp=t, maskmiddle=f, ambiguous=all). A spacer is present when it is found in at least the minimum count of reads (contigs for assemblies). Known variants of standard spacers, too different to be found as the spacer (more than 1 mismatch) but named as present in the spoligotype databases, are searched the same way and counted for their spacer (e.g. spacer 3 of M. orygis, 2 mismatches): the sample page says so. The fraction of MTBC reads is the depth of the control regions divided by the depth expected from the number of bases.

Run information

Operator	Example
User	example
Computer	example
Operating system	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Started	2026-10-02 13:15:04 EDT
Finished	2026-10-02 13:15:16 EDT
Duration	12.8 s
Working directory	/example
Command	spoligotyper -i data/ -o results/ -t 8 --operator Example

Parameters

Input	/example/data
Output folder	/example/results
Minimum count	default (5 for fastq, 1 for fasta)
Threads	8
Jobs	1
Seal memory	1g
Species and lineage	yes

Software

spoligotyper	0.9.0
Python	3.14.7 (/opt/conda/bin/python3.14)
Seal	/opt/conda/bin/seal.sh
BBTools	40.02
Java	openjdk version "25.0.2-internal" 2026-01-20

Reference data

Spoligotype database	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligotype_db.txt 1,976 patterns · MD5 0a88f7b89f746de2a11e1b4af0151e36
Spacer sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligo_spacers.fasta 43 spacers · MD5 021ac8a5ba5bbfd21c21724973b699d5
Spacer variants	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spacer_variants.fasta 1 known variant(s) · MD5 2107b897bc759ce9d96a5158a9909f3f
Species markers	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/markers.fasta MD5 87135cd99f26936911997867c6f1964b
Lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_barcode.tsv MD5 045f6121a83eeff67068ad736e6debc6
Lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_snps.fasta MD5 7e662a9baee450e79065705eb7b79021
Livestock lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/livestock_barcode.tsv MD5 042f0761e4ac34e498c99653f514772b
Livestock lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/livestock_snps.fasta MD5 8f8ea8ccbaacf13349f67fbe2cce5edb
Lineage 1 sublineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/l1_barcode.tsv MD5 455535414467135457fbf540270491f9
Lineage 1 sublineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/l1_snps.fasta MD5 ededd894dffde9e792e677b3fe6b3baa
SIT database	/opt/spoligotyper/sit_database.tsv 9,656 patterns, 3,850 SITs · SHA-256 4b17cb26525d2de4bb7a367350fba020199c2e4ea03f9254bb7da488b5828b08 SIT database for spoligotyper, converted from SpolLineages Spoligo_list.csv (SITVIT2 patterns)

References

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